

Forensic Copy Verification using SHA-256

Python Code

```
import hashlib def sha256_of(filename): with open(filename, "rb") as f: return  
hashlib.sha256(f.read()).hexdigest() with open("original_disk.img", "wb") as f: f.write(b"HEADER" +  
bytes(range(256)) * 4 + b"FOOTER") with open("original_disk.img", "rb") as src: with  
open("forensic_copy.img", "wb") as dst: dst.write(src.read()) original_hash =  
sha256_of("original_disk.img") copy_hash = sha256_of("forensic_copy.img") print("Original image  
hash:", original_hash) print("Forensic copy hash :", copy_hash) if original_hash == copy_hash:  
print("Match: VERIFIED - exact bit-stream copy") else: print("MISMATCH - copy corrupted")
```

Sample Output

Original image hash:
d50630bee2bf9926344781042d10cb09377730818efda8e80f9eeec21008a9b0 Forensic copy hash
: d50630bee2bf9926344781042d10cb09377730818efda8e80f9eeec21008a9b0 Match: VERIFIED -
exact bit-stream copy